

TADESSE JEMAL

FULL-STACK WEB DEVELOPER

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SUMMARY

Results-Driven Full-Stack Developer with **5+ years of experience** in designing, developing, and deploying high-performance web applications. Expertise in **JavaScript, Python, and Ruby**, and **LLM-powered** AI systems, with advanced proficiency in **React, Redux, Rails, Node.js, and modern AI integration workflows**. Adept at delivering **scalable, user-centric solutions** within strict deadlines and budget constraints. Strong problem-solving skills with a commitment to **clean, maintainable code** and **best practices**. Experienced in integrating AI automation, prompt engineering, and intelligent data-processing systems to enhance product efficiency and user experience. Thrives in **collaborative Agile teams** and **remote work environments**, with a passion for continuous learning and adopting emerging technologies.

SKILLS

Front-End Development: HTML, CSS3, JavaScript, React JS, Vue.JS, Next JS, TypeScript, Redux

Styling: CSS3, SASS, Bootstrap, Tailwind CSS, Material UI

JavaScript Data Visualization Libraries: D3.js, Chart.js, Highcharts, Three.js

Back-end development: Python, Flask, Ruby, NodeJS, Ruby on Rails

Tools & Methods: Git, GitHub, Jest, Respect, TDD, Chrome, Redux DevTools, CI/CD

Database: PostgreSQL, MySQL, MongoDB, SQLite

Professional: Remote Pair-Programming, Teamwork, Mentoring, Communication Skill

Cloud & Hosting Service: Render, Netlify, Vercel, Railway, Heroku, AWS

EXPERIENCE

HIGHLIGHTED PERSONAL PROJECTS

Full-Stack Developer and AI Specialist

Remote
September 2020 – Present

Space-Travelers-Hub

- Designed and developed a **commercial and scientific space travel platform** enabling users to browse and book missions.
- Built a responsive front-end with **React, Redux, and modern CSS3**, ensuring seamless UI/UX.
- Implemented state management with **Redux** for efficient data flow and real-time updates.

- Deployed a scalable architecture with **integrated APIs** for dynamic content rendering.
- [Live link](#)

Hotel Booking App

- Built a responsive hotel reservation platform integrating an external API to display available rooms.
- Enabled secure user authentication and booking functionality.
- Utilized **React, Tailwind CSS, Redux, Rails, and PostgreSQL** for full-stack development.
[Confidential – Details Available Upon Request]

School Management System

- Architected and delivered a full-stack academic management platform to streamline enrollment, scheduling, attendance, grading, and cross-department operations.
- Implemented **AI integration** to automate repetitive administrative tasks, provide intelligent assistance, and support faster data-driven decisions for school staff.
- Developed secure, role-based authentication and access control for Super Admins, Branch Admins, Teachers, Finance Officers, Librarians, Parents, and Students.
- Built with **Next.js 15, TypeScript, React, Tailwind CSS 4, NextAuth v5, Prisma ORM, and PostgreSQL**. **[Confidential – Details Available Upon Request]**

AI-Driven Health System Management Platform

- Spearheaded the design and delivery of an intelligent, full-stack healthcare platform that unified patient registration, appointment orchestration, EHR workflows, billing, and cross-department coordination.
- Integrated **AI capabilities** to automate repetitive administrative processes, provide real-time staff assistance, and accelerate data-informed clinical and operational decisions.
- Implemented secure, role-based access control for Super Admins, Branch Admins, Doctors, Nurses, Lab Technicians, Pharmacists, Accountants, Receptionists, and Patients.
- Built with **Next.js 15, TypeScript, React, Tailwind CSS 4, NextAuth v5, Prisma ORM, and PostgreSQL**. **[Confidential – Details Available Upon Request]**

MICROVERSE Mentor (Volunteer)

Remote
August 2022 – Present

- Mentored junior web developers, providing technical support through code reviews, enhancing code quality and performance.
- Offers strategies for maintaining motivation and managing project deadlines, essential for long-term success in the program.

CNET Software Technologies PLC
Front End Developer

Addis Ababa, Ethiopia
January 2020 – December 2021

- Conceptualized design ideas through the use of storyboards, process flows, and sitemaps to illustrate clear navigation paths.
- Designed and developed five websites utilizing React, focusing on user-friendly interfaces and responsive design.
- Engineered two taxi appointment applications tailored for end-user efficiency, enhancing the overall booking experience.

Technologies: React.js, JavaScript, HTML5, CSS3, Bootstrap, REST APIs, Git, Responsive Design, UI/UX Design

DoWell Research India
React Developer

Singapore (Remote)
May 2022 – November 2023

- Developed scalable, reusable front-end components and libraries, significantly improving development efficiency and code maintainability across projects.
- Translated complex UI/UX designs into high-performance, production-ready code, ensuring cross-browser compatibility and responsiveness on a wide range of devices.
- Collaborated closely with back-end engineers to integrate RESTful APIs and deliver seamless user experiences.
- Engineered advanced features such as voice and video recording tools and dynamic PDF/invoice generation using automated template systems.
- Contributed to overall product stability, performance, and scalability through clean architecture and continuous code optimization.

Technologies: React.js, JavaScript, TypeScript, Redux, HTML5, CSS3, Tailwind CSS, REST APIs, PDF Generation, Git, Responsive Design

Entro
Full Stack Developer

Reykjavík, Iceland (Remote)
November 2023 - May 2026

- Translated Figma designs into responsive, accessible web pages, ensuring adherence to modern web standards and WCAG accessibility guidelines.
- Built RESTful APIs using Python Flask and designed relational database models to seamlessly integrate front-end and back-end operations.
- Collaborated with the team to engineer a door automation system using Python, enabling secure access control based on verified bookings.
- Collaborated with the team to develop an MQTT-based real-time communication system for live data transfer between the booking platform and IoT devices.

- Designed and implemented a professional **admin dashboard** with granular control over bookings, access logs, and system settings, including comprehensive **user management features**.
- Played a key role in ensuring the reliability, security, and scalability of the system in a real-world environment.

Technologies: React.js, Python, Flask, PostgreSQL, MQTT, Node.js, REST APIs, IoT Systems, Tailwind CSS, Figma, Git, Docker

Parka Ehf
Full Stack Engineer

Reykjavík, Iceland(Remote)
July 2025 – March 2026

- Developed responsive and accessible web interfaces from design mockups, ensuring compliance with modern web standards and WCAG accessibility guidelines.
- Built RESTful APIs and backend services using **Node.js** and Python Flask, integrating them with relational databases for smooth front-end and back-end communication.
- Collaborated on **door automation systems, carwash management, and parking solutions**, enabling secure access control and automated service workflows.
- Engineered a **real-time MQTT communication system** for live data transfer between IoT devices and the booking platform.
- Designed and implemented a comprehensive **admin dashboard** with control over bookings, carwash and parking sessions, access logs, system settings, Monitoring, and user management.
- Focused on improving the **reliability, security, and scalability** of the system in production, supporting multiple service types simultaneously.

Technologies: React.js, Node.js, Python Flask, PostgreSQL, MQTT, Express.js, IoT Systems, Tailwind CSS, REST APIs, Docker, Git

Addis Ababa Health Bureau
Software Developer (Contract)

Addis Ababa, Ethiopia
Jan 2024 – November 2025

- Engineered and maintained a comprehensive, full-automation health management system coordinating patient intake, triage, doctor/pharmacist workflows, laboratories, and bed management.
- Developed the system's front-end using React.js, creating an intuitive and unified interface for all clinical and administrative roles to interact with the integrated platform.
- Built secure, high-performance backend services and RESTful APIs using Python and Django to orchestrate complex data flow across the entire patient care continuum.
- Integrated multiple AI assistance modules, including a diagnostic support tool for doctors, a patient-facing chatbot for triage and FAQ, and systems for generating higher-level decision reports from aggregated clinical data.
- Collaborated directly with healthcare professionals to translate complex clinical and operational requirements into clean, scalable, and compliant software features.

- Designed and optimized robust PostgreSQL database schemas and data models to handle relational data for patients, staff, medical records, inventory, and real-time system status.

Technologies: React.js, Python, Django, PostgreSQL, REST APIs, AI Integration, Chatbot Systems, Healthcare Systems, Git, Docker

Valentine Advisors
Lead Software Engineer

San Francisco Bay Area, USA (Remote)
March 2025 – present

- Built and maintained an AI-powered media monitoring and reporting system for large-scale media analysis and automation workflows.
- Developed LLM-based article analysis pipelines using prompt engineering to dynamically analyze media data for different companies and industries.
- Processed and analyzed more than 100,000+ media articles from LexisNexis and other publication data sources.
- Developed backend services and API integrations using Python and FastAPI for AI workflows, reporting systems, and external integrations.
- Designed and managed PostgreSQL/Supabase databases for storing media articles, AI analysis results, prompts, and reporting data.
- Implemented parallel batch-processing systems to improve AI analysis speed and large-scale data processing performance.
- Built React.js dashboards for workflow management, publication analysis, reporting visualization, and frontend-backend interaction.
- Created dynamic reporting systems that generated automated AI-driven media insights and analysis reports.
- Integrated Google-based automation workflows and external services into the reporting and analysis pipeline.
- Managed production deployment and infrastructure on Railway for scalable AI processing and reporting services.
- Worked on AI automation systems, prompt optimization, backend integrations, and scalable data analysis workflows across the platform.

Technologies: Python, FastAPI, React.js, PostgreSQL, Supabase, OpenAI API, Gemini AI, Prompt Engineering, Railway, REST APIs, Async Processing, AI Automation, Docker, GitHub Actions, LexisNexis Integration

Entro Ethiopia Software & Hardware Development PLC
Technical Project Manager | AI & Full-Stack Engineer

Addis Ababa, Ethiopia
Jan 2025 – Present

- **Product & Project Leadership** – Define product vision, manage agile teams, and deliver scalable software solutions.
- **AI Engineering** – Build and integrate LLM-based systems (OpenAI, Gemini) for automation and intelligent applications.

- **Full-Stack Development** – Develop modern web applications using React.js, Node.js, and Python frameworks.
- **System Architecture** – Design scalable backend systems, APIs, and distributed architectures.
- **Database Design** – Build and optimize PostgreSQL/Supabase databases for high-performance applications.
- **DevOps & Deployment** – Manage CI/CD pipelines, Dockerized applications, and cloud deployments.
- **Team & Stakeholder Management** – Lead cross-functional teams and coordinate technical delivery.

Technologies: React.js, Node.js, Python (FastAPI, Flask, Django), PostgreSQL, Supabase, OpenAI API, Gemini AI, Docker, GitHub Actions, REST APIs, CI/CD, AWS/Railway, Microservices Architecture, Agile/Scrum

EDUCATION

MICROVERSE

July 2021–October 2022

Remote Full Stack Web Development Program, Full Time

Completed an intensive 1-year remote program focused on full-stack web development.

- Mastered core computer science concepts including algorithms, data structures, and object-oriented programming.
- Gained advanced skills in **front-end development, back-end development, and database management** through real-world projects and collaborative development.
- Accumulated over 1800 hours of hands-on experience using technologies such as JavaScript, React, Ruby on Rails, and PostgreSQL.
- Developed strong capabilities in remote pair programming, version control (Git/GitHub), agile workflows, and team communication using daily standups and code reviews.

BAHIR DAR UNIVERSITY

September 2016–June 2020

BSC in Computer Science

- Participated actively in group projects, applying theoretical knowledge to practical scenarios.
- Developed a comprehensive understanding of programming languages such as JavaScript, C++, and Python.
- Completed a capstone project titled “Smart Inventory Management System,” using Python for backend and JavaScript for frontend.